

Element A: Presentation and Justification of the Problem

Problem Statement and Details (A1, A2, A5)

The people and animals involved in this problem are the students (us), Mr. Roberson, the wildlife center staff, staff at Duke, and the rabbits we are building the tunnel for.

They are planning and designing a new cage for the rabbits, that is more suitable for them and is safer. We are also making the cage easier for workers to be able to clean the cage. The rabbits will have easier access to the cages, tunnels, getting out, etc.. The situation looks unsafe for the rabbits with sharp brackets in the way, not staff or rabbit friendly, and has a hard time being cleaned as well as the rabbits not able to be seen. There need to be more ways to get the rabbits out as well. This is not a major issue, but it is something that can be improved for the rabbits safety, and even for the workers own benefits. It is definitely something that needs changes added to it to make the cage better. It is a big enough issue that it needs to be changed. The main problem is to provide a safer habitat for the rabbits, that is more accessible for the caretakers to clean and to retrieve the rabbits. The problem can occur when rabbits need to be retrieved out of their cage for emergency reasons, or other reasons, and the cage could injure the rabbits because of sharp brackets, or the rabbits cannot be seen, so the caretakers cannot therefore get the rabbits out of the cage. The problem happens because of how it was built and the material used in the cage and the sharp brackets inside the cage. The cage needs to be rabbit safe with non toxic materials because they like to chew on their enclosure. With Sanctuaries and Rabbit owners this problem can occur quite often based on the area, environment, and the rabbit itself. All these key facts help determine what type of cage the rabbits need. Its not a big issue but it could be common.

Stakeholder Concerns and Connections (A3, A4)

The Primary stakeholder of this project is the wildlife center staff. Concerns that these stakeholders have are they want to have a safe cage that is non toxic, safe for the rabbits to play on that have no sharp edges, be able to be seen by the workers, easy to clean, able to take apart and put back together easily/ easy to transport, and easy access to the rabbits.

Problem Importance (A6)

This problem needs to be solved immediately because the rabbits are already in harm and we do not want them to be in harm longer than they should. The longer the rabbits are at harm the more likely they are to become hurt and could even die due to some problems being shown with the cage that they currently have. It is so important that the rabbits have a safe environment and the wildlife center can trust the structure the rabbits are in. The rabbits need to have a non toxic environment and an overall safe environment, although the one they are in right now is the total opposite of safe. The cage the rabbits are currently in more than likely contains unhealthy wood for the rabbits to consume, has sharp edges, blinds the workers of what the rabbits are doing, and is not easy to clean or move around. Statistics show that rabbits are prone to biting their cages,

<https://www.sciencedirect.com/science/article/abs/pii/S0168159196011410> this study shows the behavior of rabbits and what things they are likely to do, and one of those things is chewing on stuff. This being said we need to really focus on making the cage sturdy along with being safe so that the rabbits do not consume any harmful chemicals from materials. The longer the rabbits are in the cage the longer they are more than likely harming themselves, while being unaware.

<https://kavee.com/blogs/the-bunny-blog/keep-your-rabbit-cage-clean?pb=0>

This study also provides us with the information that it can take up to 15 minutes give or take to clean a rabbit cage. If the cage can have some features to make it easier to take apart then that time should reduce significantly. In conclusion the rabbits need to have a new cage as soon as possible to keep them safe from harming themselves any further than they may have already been. This problem also needs to be solved fast to make the workers' life easier, so they can be able to work without feeling as if they have to go and open their rabbit's cage 24/7 to make sure they are okay.